AI SMART WASTE MANAGEMENT SYSTEM FOR SMART CITIES

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PROBLEM STATEMENT

Urban areas generate large amounts of waste due to rapid population growth.
Traditional waste collection follows fixed schedules without knowing bin fill levels.
This leads to overflowing bins, pollution, health risks, and inefficient use of resources.

There is a need for a cloud-based intelligent system using Microsoft Azure.

PROPOSED SYSTEM / SOLUTION

An Azure-based AI Smart Waste Management System is proposed.
The system predicts garbage bin fill levels using machine learning.
It helps municipal authorities plan waste collection efficiently.
It improves cleanliness and public hygiene in society.

SYSTEM DEVELOPMENT APPROACH (TECHNOLOGY USED)

• Azure Blob Storage – Stores waste data
• Azure Machine Learning – Trains prediction model
• Azure SQL Database – Stores structured information

• Azure App Service – Hosts monitoring dashboard
• Azure Monitor – Tracks system performance

Algorithm Used: Random Forest Regression

• Uses historical waste data for training
• Handles non-linear data effectively
• Provides accurate prediction of bin fill levels

DEPLOYMENT

The model is trained in Azure Machine Learning.
It is deployed as an Azure ML endpoint.
The dashboard is hosted using Azure App Service.
This is a cloud-based, scalable, and reliable solution.

RESULT

• Accurate prediction of garbage bin fill levels
• Reduction in overflowing bins
• Optimized waste collection schedules
• Improved cleanliness in urban areas

CONCLUSION

The project demonstrates the effective use of Microsoft Azure and Artificial Intelligence.
It solves a real-world societal problem.
It improves efficiency, hygiene, and resource utilization.
It supports smart city initiatives.

FUTURE SCOPE

- Integration with Azure IoT Hub for real-time sensors
- Mobile application for sanitation workers
- Expansion to multiple cities
- Advanced AI and deep learning models

REFERENCES

- Microsoft Azure Documentation
- Azure Machine Learning Documentation
- Research papers on Smart Waste Management

GITHUB LINK

https://github.com/Harika-23-27/Smart_waste_management